From Rewrite Semantics to Evidence: A Conceptual Primer Linking GSLT, OSLF, Behavioral Types, World-Model Evidence, and Knuth–Skilling Measurement

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Abstract

This note distills a conceptual architecture for connecting (i) operational/rewrite semantics of languages, (ii) an *intrinsic* logic of behavioral properties and "native types" derived from that semantics, (iii) an observer- and evidence-aware world-model semantics, and (iv) quantitative measurement principles for plausibility/evidence that justify when scalar probabilities are appropriate and when richer (partial-order, multi-dimensional, or imprecise) valuations are structurally necessary. The aim is to serve as a didactic primer for future discussions and formalization work, while avoiding overclaims: each layer has its own truth conditions, computational constraints, and appropriate "claim hygiene."

Status note. Implementation-status and count claims in this draft are snapshot-scoped; verify against current repository trackers and build outputs before citing as current state. Lean-proven claims are only those tied to explicit Lean file/theorem references; broader mathematical or historical claims are literature-backed context and not asserted as Lean-proven unless explicitly stated.

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1 Executive summary: the three (plus one) layers

The unifying picture has three main layers (plus an optional fourth):

1. **Behavior layer (operational semantics).** A language is presented as a rewrite/reduction system. States are terms/configurations; steps are reductions. This is where "what can happen" is defined.

2. **Behavioral logic/type layer (OSLF-style).** From a reduction relation, one constructs a logical space of predicates on states (often organized as a Heyting algebra), together with modal operators induced by the reduction structure. This is a *behavioral* semantics: it talks about reachability, invariants, and interaction structure, not about denotations in a separate model.

3. **Measurement layer (world-model evidence + valuation theory).** A world model assigns *evidence* to queries about behavior. Evidence lives in a structured algebra (often richer than $\mathbb{R}$), supporting revision/combination. Quantitative plausibility constraints (Knuth–Skilling style) explain when and why regraduation to a scalar probability-like scale is justified, and when it is not.

4. **(Optional) Observer/indistinguishability layer (distinctions).** One may explicitly model an observer's (in)ability to distinguish states, yielding graphs or relations of indistinguishability and entropy-like summaries of the observer's discrimination. This layer is useful for resource-bounded agents and imprecise evidence.

   A core methodological stance is:

   > *Keep the behavioral logic honest and compositional; keep the measurement layer explicit about assumptions; avoid "smuggling" an oracle or a scalar collapse into the semantics without stating it.*

2 GSLT-style language specifications: rewrite systems with premises

2.1 Why a rewrite-first presentation?

Many systems of interest—programming languages, process calculi, grammar formalisms, inference engines—admit a presentation where the *dynamics* is given by rewriting:

- a syntax of terms or configurations (often multi-sorted),
- equations / structural congruence (rearrangements that preserve meaning),
- rewrite rules (directed dynamics),
- *premises* (side conditions) such as freshness, matching constraints, or oracle relations (e.g., a grounded/native function call).

This style is attractive because it provides a single substrate for:

1. an executable engine (compute next steps),
2. a declarative relation (what *counts* as a step),
3. and higher-level constructions (modalities, type systems, evidence semantics).

2.2 Premise-aware rewriting is not optional

Realistic specifications frequently require side conditions: fresh-name generation, matching constraints, or external computations. A key distinction is:

- **Internal premises:** decidable properties computed on syntax (freshness, equality, matchability).
- **Oracle premises:** relations answered by an external environment (grounded calls, database queries, probabilistic sensors).

A premise-aware rewrite semantics should be defined in *two* aligned ways:

1. a **declarative** step relation $p \Rightarrow q$ that explicitly quantifies over premise witnesses;
2. an **executable** step function that computes candidate reducts and is proved sound (every computed reduct is valid) and complete (every valid reduct is computed), relative to the premise environment.

This "engine $\leftrightarrow$ declarative" equivalence becomes a central lemma reused throughout the stack.

3 OSLF: behavioral logic from operational semantics

3.1 The minimal construction: modalities from a reduction relation

Let S be a set (or type) of states and let $R \subseteq S \times S$ be a one-step reduction relation. A *process predicate* is a subset $\varphi \subseteq S$ (or a function $S \to \mathsf{Prop}$).

Define modal operators on predicates by:

$$(\Diamond\varphi)(p) :\Longleftrightarrow \exists q,\ R(p,q) \ \wedge\ \varphi(q),$$
$$(\Box\psi)(p) :\Longleftrightarrow \forall q,\ R(p,q) \to \psi(q).$$

Principle 1 (Galois connection)**.** Under these definitions, $\Diamond$ and $\Box$ form a Galois connection:

$$\Diamond\varphi \subseteq \psi \quad \text{iff} \quad \varphi \subseteq \Box\psi.$$

This single adjunction is the seed of a large amount of structure: it yields monotonicity, distributive properties, and a disciplined language of reachability and invariance claims.

3.2 From predicates to a Heyting algebra of "types"

Predicates ordered by implication/inclusion form a complete lattice (under mild assumptions). With implication interpreted as Heyting implication, one obtains a Heyting algebra:

- $\varphi \wedge \psi$ = conjunction (intersection),
- $\varphi \vee \psi$ = disjunction (union),
- $\varphi \Rightarrow \psi$ = Heyting implication (weakest predicate making φ entail ψ),
- $\top$ and $\bot$ = tautology and contradiction.

Remark 1. This is a *logic of behavior*: propositions are predicates on states. It is not, by itself, a denotational model assigning external meanings to programs.

3.3 The topos upgrade: subobject classifiers and presheaf semantics

The "predicate = subset" picture is Set-level. A more categorical presentation treats predicates as *subobjects* and organizes them via the *subobject fibration*. For presheaf categories $\widehat{C} \simeq C^{op} \to \mathsf{Type}$, predicates correspond to subpresheaves, and the subobject classifier Ω is described by sieves.

The conceptual steps (useful as a formalization roadmap) are:

1. construct Ω in $\widehat{C}$ (sieves object),
2. show the subobject presheaf $\mathrm{Sub}(-)$ is representable by Ω,
3. obtain **HasClassifier** for the presheaf topos constructively,
4. define change-of-base along morphisms and derive $\exists_f \dashv f^* \dashv \forall_f$,
5. prove Beck–Chevalley conditions for pullback squares,
6. instantiate the OSLF "modalities from a span" story internally in the topos/hyperdoctrine.

Remark 2. In practice, it is helpful to keep a clear separation between: (i) the constructive presheaf/topos proofs (classifier, adjoints, Beck–Chevalley), and (ii) the operational reduction engine and its soundness/completeness. The full story requires both.

4 Behavioral ontology: bisimulation, logical equivalence, and hypotheses

4.1 Bisimulation as "semantic indistinguishability"

Given a transition system (S, R), a (strong) bisimulation is a relation $\sim$ such that $p \sim q$ implies matching step structure in both directions. Bisimulation is typically an equivalence relation, enabling the quotient $S/{\sim}$. Intuitively:

Bisimulation classes are candidates for "behavioral individuals."

4.2 Modal logics and characterization theorems

Modal formulas built from $\Diamond$, $\Box$, and connectives are usually *bisimulation-invariant*: if $p \sim q$ then p and q satisfy the same modal formulas. A deeper question is *completeness*: does "satisfies the same formulas" imply bisimulation?

A classical phenomenon (often associated with Hennessy–Milner style results) is:

- Under **image-finiteness** (finite branching) or similar hypotheses, a finitary modal logic can characterize bisimulation.
- Without such hypotheses, one often needs **infinitary** conjunction/disjunction and/or **μ-calculus** fixpoints to fully characterize bisimulation.

Remark 3 (Philosophical reading)**.** Image-finiteness can be read as a *bounded-choice* assumption: finite tests can in principle separate behaviors because there is not an infinitary fan-out at each step. When branching is unbounded or behaviors are inherently recursive/reflective, finitary observation is not complete without enrichment (infinitary or fixpoint structure).

5 World-model semantics: evidence, revision, and guarded rewrites

5.1 Why a world model?

OSLF gives a compositional logic of behavioral predicates. But an agent typically needs more than boolean truth: it needs *graded evidence* under uncertainty, partial observability, and correlation.

A world model provides:

- a state space W (often containing a joint distribution or structured uncertainty),
- a query language Q (questions about the world or computation),
- an evidence structure E,
- an evidence extraction map $\mathsf{evidence} : W \times Q \to E$,
- a revision/combination operation on world states or evidences.

5.2 Evidence quantales and Heyting-valued semantics

A common pattern is to let E be a (commutative) quantale or related algebra: a partially ordered space with combination and joins, supporting both aggregation and revision. In Heyting-compatible settings, connectives are interpreted by the lattice/Heyting operations.

Principle 2 (No premature scalar collapse)**.** If evidence is naturally partial-order valued (multi-dimensional, interval/imprecise, or structured), collapsing to a single real number without stated assumptions generally loses information. Scalar probabilities arise as a *view* under additional constraints.

5.3 Guarded rewrite calculus for queries

A practical and honest approach is to use a rewrite calculus for queries with explicit side conditions Σ:

Rules are applied only when stated assumptions about independence, approximation, or model class hold.

This enables:

- derivations that are sound relative to the world model and side conditions,
- explicit tracking of approximation,
- modular composition of inference heuristics with a small trusted kernel.

6 Knuth–Skilling measurement: when do numbers appear?

6.1 Two complementary roles of KS-style theory

Knuth–Skilling style axiomatizations are useful in two ways:

1. **Measurement value theory:** if plausibility values have certain order/combination symmetries, there exists a regraduation into a real scale where combination becomes additive (or otherwise tractable).
2. **Valuation on event/proposition lattices:** if events form a (distributive/Boolean) lattice, then plausible axioms (sum rules, chain rules) yield probability-like calculus, including Bayes-like laws under suitable structure.

6.2 Totality vs. imprecision

A recurring technical and philosophical distinction is:

- **Total/linear order assumptions** enable faithful embedding into $\mathbb{R}$ and classical probability-like behavior.
- **Partial orders** naturally support imprecise or multi-dimensional evidence; representation by a single scalar is generally not faithful without loss.

In the integrated picture, one uses KS-style results to justify *which* scalar views are legitimate, and to document the assumptions needed for each view.

7 Distinction graphs: observer-relative indistinguishability and graphtropy

7.1 From equivalence relations to graphs

If indistinguishability is transitive, it forms an equivalence relation and yields a partition. If indistinguishability is not transitive (as with resource-bounded or noisy observers), it is naturally represented as a graph of non-distinction, possibly weighted. This complements bisimulation:

- **Bisimulation** is an "ideal observer" equivalence induced by the semantics.
- **Distinction graphs** model an actual observer's discrimination regime (tests, thresholds, noise).

7.2 Graphtropy as measurement of discrimination

Graph-based generalizations of logical entropy ("graphtropy") summarize how much an observer distinguishes, and weighted versions can incorporate graded distinguishability. This provides a bridge between qualitative behavioral ontology and quantitative measurement, especially when scalar probability is not the right primitive.

8 Integration blueprint: a clean end-to-end story

8.1 A canonical pipeline

A disciplined, non-overclaimed pipeline looks like:

1. **Language spec (GSLT)**: syntax + equations + rewrite rules with premises.
2. **Operational engine**: executable one-step search (premise-aware), with soundness/completeness to a declarative step relation.
3. **Behavioral logic (OSLF)**:
   - predicate space (Heyting structure),
   - modalities induced by reduction (Galois connection),
   - (optional) topos/hyperdoctrine internalization.
4. **World model**: evidence-valued semantics for atoms/queries; revision rules.
5. **Measurement constraints**: KS-style regraduation/valuation theorems justify scalar views and clarify when imprecision is required.
6. **Tools**: bounded checkers, abstract interpreters, and executable analyzers as approximations with stated correctness guarantees.

8.2 Two-track R&D that meets in the middle

Two complementary development tracks help avoid both "math-only" and "tool-only" traps:

Track A: principled math-first Prioritize constructive topos/hyperdoctrine foundations: classifier, adjoints, Beck–Chevalley, and internal change-of-base. This ensures the derived modalities and quantifiers are not merely asserted but constructed.

Track B: executable checker-first Prioritize premise-aware rewriting engines, bounded checkers, and agreement tests. Prove soundness (and completeness where feasible) so the executable tool is not a black box.

Where they meet The meeting point is an end-to-end theorem: "checker says `sat` $\Rightarrow$ semantic satisfaction," followed by transport into world-model evidence semantics (possibly via thresholding or an evidence-to-Prop view).

9 MeTTa and GF as motivating clients

9.1 MeTTa as a GSLT: state machines + oracle boundaries

A key modeling lesson is to treat a realistic language like MeTTa as a *state transition system*:

- states include plan/stack, atomspace, bindings, and intermediate results;
- rewriting encodes one machine step;
- grounded/native calls are handled as explicit oracle premises.

This preserves honesty: the semantics does not pretend to compute what must be queried.

9.2 GF as a query/meaning interface

Grammar formalisms can be treated as rewrite/derivation systems producing structured meanings and paraphrases. This creates opportunities to:

- express semantic queries as OSLF formulas over derivation dynamics,
- interpret queries evidentially in a world model,
- compare linguistic variants by behavioral invariants (language-independent abstract structures).

10 Claim hygiene: avoiding facepalmable statements

Principle 3 (Do not conflate layers)**.** Avoid claims of the form "we derived X" when what is implemented is:

- hand-defined objects + proofs they satisfy an abstract interface, rather than
- an executable algorithm producing those objects from the language spec.

Both are valuable, but they are different claims.

Principle 4 (Make oracle boundaries explicit)**.** If grounded calls, external databases, or probabilistic sensors are involved, keep them as explicit premises/oracles with a clearly stated interface and semantics.

Principle 5 (Document assumptions for scalar views)**.** If a scalar probability/plausibility scale is used, document the order/separation/regularity assumptions and present scalarization as a *view* on richer evidence unless proven otherwise.

11 A minimal "next theorem chain" to target

The following chain is a compact target for a rigorous end-to-end bridge:

Target Theorem 1 (Engine–declarative equivalence)**.** Premise-aware executable reduction is sound and complete for the declarative relation.

Target Theorem 2 (OSLF adjunction)**.** Modal operators induced from the reduction relation satisfy $\Diamond \dashv \Box$.

Target Theorem 3 (Topos foundations (constructive))**.** In the presheaf topos: Ω exists, $\mathrm{Sub}(-)$ is representable, change-of-base has $\exists \dashv f^* \dashv \forall$, and Beck–Chevalley holds for pullbacks.

Target Theorem 4 (Checker soundness)**.** If the bounded checker returns `sat`, then semantic satisfaction holds.

Target Theorem 5 (Evidence semantics bridge)**.** Define an evidence-valued interpretation of atoms and lift it to formulas. Show checker soundness implies a thresholded (or otherwise structured) evidence judgment.

Target Theorem 6 (Measurement justification)**.** Under explicit KS-style assumptions, prove a regraduation/representation theorem showing when scalar probability-like semantics is a conservative view.

Acknowledgements and scope

This note is intentionally conceptual and claim-hygienic. It is not a substitute for the detailed formal developments in a proof assistant, nor does it assert that every item above is already implemented. Its purpose is to provide a shared map for future formalization and discussion.

Selected references (non-exhaustive)

- S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic.*
- P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium.*
- B. Jacobs, *Categorical Logic and Type Theory.*
- M. Hennessy and R. Milner, work on modal logic and bisimulation (Hennessy–Milner theorem phenomenon).
- Knuth–Skilling style work on measurement foundations of inference (regraduation, valuation theory).
- Work on subobject classifiers and presheaf toposes; see also formalization libraries (mathlib) for constructive developments.
- Work on logical entropy (Ellerman) and observer-relative distinction frameworks (distinction graphs, graphtropy).